

AM CVn: Testing a Superhump Interpretation

TODO: Team Members

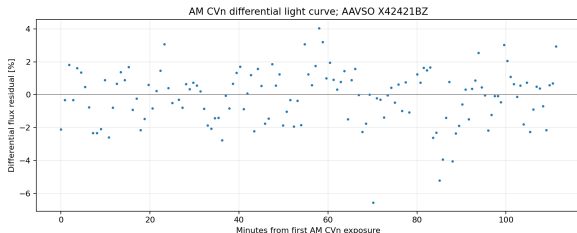
PHYSICS 100 Final Project

Scientific Goal

- AM CVn is an ultracompact helium-accreting binary.
- Literature periods: orbital 1028.7322 s, positive superhump 1051.2 s, strong harmonic near 525.6 s.
- Period sources: Skillman et al. 1999; Nelemans et al. 2001; Roelofs et al. 2006.
- Question: can our 2026 May 20 photometry tighten those constraints?
- Rule: separate free-period constraints from literature-conditioned fits.

Observations

- 140 V-band frames.
- 35, 40, and 45 s exposures.
- 138 good points after rejecting two artifact frames.
- Baseline: 111.20 min.
- Covers 6.35 cycles of a 1051.2 s superhump.

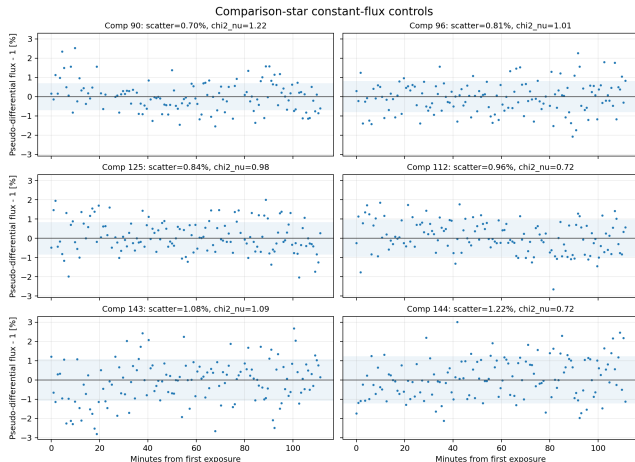


Reduction

- Bias, dark, and star-masked flat-field calibration.
- AAVSO sequence X42421BZ used for target and comparison-star geometry.
- Fluxes converted to rates before combining different exposure times.
- Comparison stars normalized by their own median before forming the ensemble.
- Robust target scatter: 1.51 percent; median formal error: 0.90 percent.

Comparison-Star Controls

- Each comparison star was treated as a pseudo-target.
- The pseudo-target was divided by the remaining comparison ensemble.
- Robust scatter: 0.70–1.22 percent.
- Empirical constant-model χ^2_ν values are near unity.
- This supports the ensemble as a stable photometric reference.

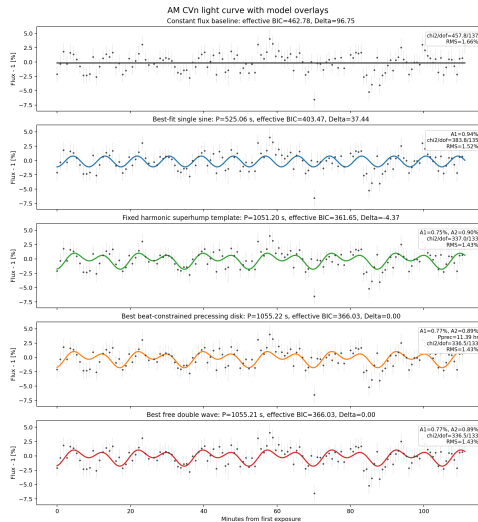


Models and BIC

- All models fit sine/cosine curves to the same light curve.
- Harmonic superhump: fix $P_{\text{sh}} = 1051.2$ s and add the $2f$ harmonic.
- Free double wave: same two-frequency shape, but P is fit from our data.
- Beat disk: subtract frequencies, $f_{\text{sh}} = f_{\text{orb}} - f_{\text{prec}}$, then fit that waveform.
- χ^2 : weighted residual size.
- $\text{BIC}_{\text{eff}} = \chi^2 + k_{\text{eff}} \ln N$: fit quality plus complexity penalty.

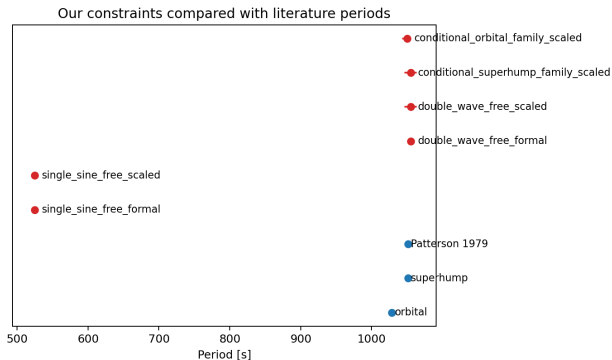
Model	Period	k_{eff}	$\Delta\text{BIC}_{\text{eff}}$
Constant flux	—	1	96.75
Single sine	525.06 s	4	37.44
1051.2 s + harmonic	fixed	5	-4.37
Beat-constrained disk	1055.22 s	6	0.00
Free double wave	1055.21 s	6	0.00

Model Fits



Constraints

- Free single sine: 525.06 s; Monte Carlo median $524.38^{+5.75}_{-3.84}$ s.
- Free double wave: $1055.21^{+13.10}_{-9.63}$ s.
- Best independent period measurement: free double wave.
- Best conditional model: fixed 1051.2 s + harmonic.
- Both support the known superhump family.



Validation

- Multi-harmonic Lomb-Scargle recovers the double-wave fundamental.
- Fixed-time Monte Carlo perturbations replace row-resampling bootstrap as the primary error analysis.
- Beat-constrained model implies $P_{\text{prec}} \approx 11.39$ hr, inferred from the beat relation rather than directly observed.
- Leave-one-comparison-star-out period span: 5.2 s.
- Outlier clipping period shift: 0.02 s.
- Even the best sine/double-wave models have reduced chi-squared around 2.5, so uncertainties are empirically scaled and conclusions are conservative.
- Residual periodogram has no significant remaining peak.
- Systematics terms lower RMS to 1.41%, but worsen BIC/CV.
- Robust losses shift the answer to 1057–1059 s and worsen RMS.
- First/second halves differ by 22.8 s.

Could It Be Something Else?

- Broadband photometry alone cannot prove the object class.
- Literature spectroscopy shows helium accretion signatures and an S-wave orbital feature.
- A non-binary pulsator would need a spectrum without accretion features and a different multi-period pattern.
- The decisive test is time-resolved spectroscopy, not another photometric periodogram.

Conclusion

- We detect AM CVn short-timescale variability.
- The 525 s single-sine signal is best interpreted as a harmonic.
- The fixed 1051.2 s harmonic template has the best effective BIC, but it is a literature-conditioned model.
- The free double wave is the best independent period measurement.
- Detrending, robust likelihoods, and comparison subsets do not improve the constraint.
- We do not tighten the literature periods with this dataset.
- A longer multi-night baseline is the path to a stronger constraint.